Simpson's non-abelian Hodge correspondence establishes a diffeomorphism between different moduli spaces of sheaves on a projective variety X over the complex numbers. Specifically, let $M^0_{n,Dol}(X)$ be the moduli space of semistable Higgs bundles of rank n and degree 0 on X and let

$$M^0_{n,B}(X) \coloneqq \{\mathrm{Hom}(\pi_1(X), GL_n)\} \,//GL_n$$

be the variety of n-dimensional representations of the fundamental group of X. Then Simpson's correspondence is a diffeomorphism

$$\varphi\colon M^0_{n,Dol}(X) \to M^0_{n,B}(X) \tag{0.1}$$

between these varieties, inducing therefore an isomorphism of their cohomology rings. Since we are considering varieties over $\mathbb{C}$, the cohomology groups are endowed with mixed Hodge structures. But the morphism φ is not algebraic and doesn't preserve the Hodge structures. Therefore, a natural question is to try and understand how φ relates the Hodge structures.

In order to understand this, it is convenient to factor the pullback φ^* into pieces that are easier to understand. First of all, the Riemann-Hilbert correspondence establishes a diffeomorphism

$$M^0_{n,B}(X) \simeq M^0_{n,DR}(X)$$

between the character variety $Rep(\pi_1(X), GL_n)$ and the De Rham moduli space, i.e. the moduli space parametrizing pairs $(\mathcal{E}, \nabla)$ where $\mathcal{E}$ is a vector bundle of rank n on X and

$$\nabla\colon \mathcal{E} \to \mathcal{E} \otimes \Omega_X$$

is a flat connection on $\mathcal{E}$.

Simpson introduced the notion of λ-connection as a way to interpolate between connections and Higgs fields. For a fixed complex parameter λ, a λ-connection on a vector bundle $\mathcal{E}$ is a $\mathbb{C}$-linear morphism

$$\nabla_\lambda\colon \mathcal{E} \to \mathcal{E} \otimes \Omega_X,$$

respecting the Leibniz rule up to the scalar λ, namely such that

$$\nabla_\lambda(f \cdot s) = f \cdot \nabla(s) + \lambda\, s \otimes \mathrm{d}f.$$

Notice that for $\lambda = 1$ we recover the notion of a connection, while for $\lambda = 0$ we get an $\mathcal{O}_X$-linear morphism $\mathcal{E} \to \mathcal{E} \otimes \Omega_X$, making $(\mathcal{E}, \nabla_0)$ into a Higgs bundle.

Let $M^0_{n,Hdg}(X)$ be the moduli space parametrizing triples $(\lambda, \mathcal{E}, \nabla)$, where λ is a complex parameter, $\mathcal{E}$ is a rank n and degree 0 vector bundle on X and ∇ is a λ-connection on X. Th multiplicative group $\mathbb{G}_m$ acts on $M^0_{n,Hdg}(X)$ by scaling of ∇. In particular, the projection $(\lambda, \mathcal{E}, \nabla) \mapsto \lambda$ defines a $\mathbb{G}_m$-equivariant morphism

$$f\colon M^0_{n,Hdg}(X) \to \mathbb{A}^1,$$

where $\mathbb{G}_m$ acts on $\mathbb{A}^1$ with weight one. The fibers of f over 0 and 1 are $M^0_{n,Dol}(X)$ and $M^0_{n,DR}(X)$ respectively, i.e. we have the following picture

$$\begin{tikzcd} M^0_{n,Dol}(X) \arrow[r, hook, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M^0_{n,Hdg}(X) \arrow[d, "f"] & M_{DR} \arrow[l, hook', "j"'] \arrow[r, "\psi", "\cong"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] & M^0_{n,B}(X) \\ \{0\} \arrow[r] & \mathbb{A}^1 & \{1\}. \arrow[l] & \end{tikzcd} \tag{0.2}$$

Using the fact that the morphism f is $\mathbb{G}_m$-equivariant and that the fibers over all $\lambda \neq 0$ are isomorphic, one can show that i and j induce isomorphisms in cohomology. They are moreover algebraic maps, so they induce an isomorphism of mixed Hodge structures on between the cohomology of $M^0_{n,Dol}(X)$ and $M_{n,DR}(X)$. Finally, we need to understand how ψ acts on the Hodge structures. In [She17], Shende approached this problem by replacing X with a simplicial scheme Δ_X obtained by a trialeftngulation. The advantage of using this approach is that the simplicial scheme thus constructed admits an algebraic version of a universal local system, which can be used to compute the hodge weights.

Notice that the problem explained so far concerns only vector bundles of degree 0. Indeed, in order for a vector bundle to admit a nontrivial flat connection, it needs to have degree 0. On the other side, positive degree vector bundles can admit nontrivial Higgs fields, so we would like to have a version of (0.1) for positive degree Higgs bundles. The purpose of this article is to study a version of (0.1) for vector bundles of nonzero degree in the case where X is a genus g curve and to apply Shende's technique to study the Hodge structure also in this case. Thanks to [Simpson], we know that in the degree d case, the character variety in (0.1) should be replaced by

$$\left\{ (A_i, B_i) \in GL_n^{2g} \;\middle|\; \prod_{i=1}^{2g} [A_i, B_i] = e^{\frac{2\pi i}{n} d} \right\} /\!/ \, GL_n. \tag{0.3}$$

Our main contribution, is to interpret this space as character variety of a stacky curve, and use this interpretation to get an analogue of the picture described in Equation (0.2) and replicate Shende's argument. Indeed, (0.3) is a connected component of the variety of representations of the group

$$\left\langle a_i, b_i, c \mid i = 1, \ldots, g, \; \prod_{i=1}^{g} [a_i, b_i] = c, \; c^n = 1 \right\rangle.$$

The key idea is that this group can be seen as the fundamental group of an orbifold curve with a singularity of order n, which algebraically corresponds to a stacky curve with a single stacky point with stabilizer μ_n. If we let $\mathcal{X}$ be such

a stacky curve, then we get the analogue of (0.2)

$$\begin{tikzcd} M^0_{n,Dol}(\mathcal{X}) \arrow[r, hook, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M^0_{n,Hdg}(\mathcal{X}) \arrow[d, "f"] & M_{DR} \arrow[l, hook', "j"'] \arrow[r, "\psi", "\cong"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] & M^0_{n,B}(\mathcal{X}) \\ \{0\} \arrow[r] & \mathbb{A}^1 & \{1\}. \arrow[l] & \end{tikzcd} \tag{0.4}$$

The stacky curve $\mathcal{X}$ admits a line bundle $\mathcal{O}(\frac{1}{n})$of fractional degree, which we can use to twist degree d vector bundles into degree 0 vector bundles. In particular, starting from a degree d vector bundle $\mathcal{E}$ on X, we can pull it bach along $\pi\colon \mathcal{X} \to X$ and get a degree d vector bundle on $\mathcal{X}$ with trivial type in the stacky point. After twisting, we get a vector bundle of degree 0 on $\mathcal{X}$ which in the stacky point p realizes the μ_n-representation

$$\rho_n^{-d} := \begin{pmatrix} \zeta_n^{-d} & & \\ & \ddots & \\ & & \zeta_n^{-d} \end{pmatrix}.$$

The described procedure lifts to an isomorphism of stacks

$$\mathcal{M}^d_n(X) \cong \mathcal{M}^d_{n,\mathrm{triv}}(X) \cong \mathcal{M}_{n,\rho_n^{-d}}(\mathcal{X}),$$

which extends to the various moduli stacks of vector bundles described above. In particular, looking at the connected component of (0.4) determined by the invariants (n, d, ρ_n^{-d}), we recover a version of (0.1) for degree d vector bundles on X.

Finally, we want to use (0.4) to compare the Hodge structures on $\mathcal{M}_{n,\rho_n^{-d}}(\mathcal{X})$ and on

$$\left\{ (A_i, B_i) \in GL_n^{2g} \;\middle|\; \prod_{i=1}^{2g} [A_i, B_i] = e^{\frac{2\pi i}{n} d} \right\} /\!/ GL_n.$$

In the case of (0.2), this was done by Shende in [She17], by replacing the complex manifold X with a simplicial scheme Δ_X obtained from a triangulation of X. The key features of Δ_X are that its fundamental group coincides with the one of X, so it can be used to construct the stack of local systems on X, and that it has trivial Hodge Structure, giving the cohomology of moduli spaces on Δ_X an easier Hodge structure. Explicitly, Shende uses an isomorphism

to study

References

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